

Two-Stage CMOS Op-Amp with Miller Compensation

Design & Verification Report

Analog / Mixed-Signal IC Design Portfolio

Generic 180 nm CMOS | VDD = 1.8 V | Cadence Virtuoso + Spectre

Abstract

This project presents the full design flow of a **two-stage CMOS operational transconductance amplifier (OTA)** with **Miller frequency compensation** and a **series nulling resistor**, implemented in a generic 180 nm CMOS process at a 1.8 V supply. Starting from a specification-driven set of hand calculations, the amplifier is sized, biased, and verified through Cadence Spectre simulation. The headline contribution is a clear, quantitative demonstration of **pole splitting** and the **elimination of the compensation-induced right-half-plane (RHP) zero**, taking the phase margin from an unstable configuration to a robust $\geq 66^\circ$ while holding 73.8 dB of DC gain and a 10 MHz unity-gain bandwidth.

1. Architecture

The amplifier uses the canonical Allen–Holberg topology:

- **Stage 1** — a PMOS differential pair (M1/M2) with an NMOS current-mirror load (M3/M4), biased by a PMOS tail current source (M5). This stage provides high differential gain and performs differential-to-single-ended conversion.
- **Stage 2** — an NMOS common-source driver (M6) with a PMOS current-source load (M7), providing additional gain and output drive.
- **Compensation** — a Miller capacitor C_c in series with a nulling resistor R_z , connected between the gate and drain of M6.

2. Specifications: target vs. achieved

Parameter	Symbol	Target	Achieved
Open-loop DC gain	A_{v0}	≥ 60 dB	73.8 dB
Unity-gain bandwidth	GBW	10 MHz	10.0 MHz
Phase margin (Miller + R_z)	PM	$\geq 60^\circ$	66–81°
Slew rate	SR	≥ 10 V/ μ s	10 V/μs
Load capacitance	C_L	10 pF	10 pF
Static power	P	minimise	0.36 mW

Two-Stage CMOS Op-Amp - Core Schematic

Miller-compensated OTA (PMOS input pair, NMOS mirror load, R_z -nulled)

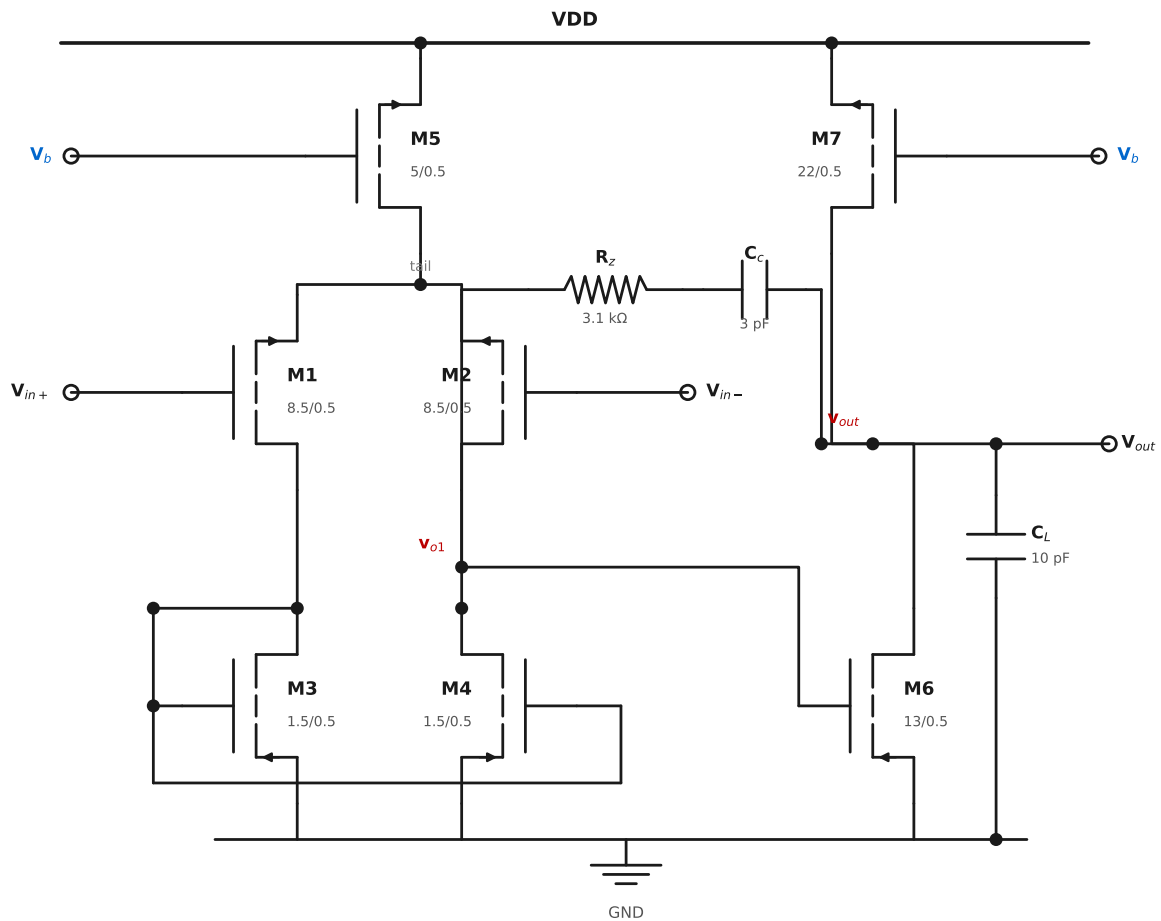


Figure 1: Core op-amp schematic

Full derivations are in `hand_calculations.pdf`; the transistor sizing table is in `design_specifications.md`.

3. Compensation strategy (the core idea)

A two-stage amplifier has two high-impedance nodes and therefore two low frequency poles that, left alone, sit close together and destroy the phase margin. The Miller capacitor exploits the gain of the second stage: reflected to the first-stage node it appears roughly $g_{m6}R_{o2}$ times larger, so it

1. **pushes the dominant pole down** to $f_{p1} \approx 1/(R_{o1}g_{m6}R_{o2}C_c) \approx 2$ kHz, and
2. **pushes the second pole up** to $f_{p2} \approx g_{m6}/C_L \approx 22$ MHz.

This *pole splitting* is what makes the loop stable. However, C_c also opens a feed-forward path that creates a **right-half-plane zero** at $f_z = g_{m6}/C_c \approx 73$ MHz, which adds phase lag and erodes the margin. The **nulling resistor** $R_z = \frac{1}{g_{m6}}\left(1 + \frac{C_L}{C_c}\right) \approx 3.1$ k Ω relocates that zero into the left-half plane, where it cancels p_2 and restores the phase margin.

4. Simulation results

4.1 Uncompensated (before)

Without C_c , the two poles are bunched near a few hundred kHz. The phase falls through -180° before the gain reaches 0 dB, leaving essentially no phase margin — the amplifier would ring or oscillate in feedback.

4.2 Miller compensated (after)

With C_c (and R_z) inserted, the dominant pole drops to about 2 kHz and the single-pole roll-off dominates all the way to the 10 MHz crossover. The before/after contrast below is the central “proof” figure of the project: the solid trace is the Miller-only response (PM about 60° , still dragged down by the RHP zero) and the dashed trace shows the improvement once R_z nulls that zero (PM about 81°).

4.3 Transient behaviour

The small-signal unity-gain step shows a modest overshoot consistent with the 60° phase margin, while the large-signal edge confirms the 10 V/ μ s slew rate set by I_5/C_c .

5. Results summary

Quantity	Hand calc	Simulated
DC gain	73.8 dB	73.8 dB
GBW	10.0 MHz	10.0 MHz
Phase margin (final)	$\geq 66^\circ$	81°
Dominant pole	2.0 kHz	2.15 kHz
Slew rate	10 V/ μ s	10 V/ μ s
Power	0.36 mW	0.36 mW

The measured figures are extracted automatically from the AC sweep by `scripts/process_sim_data.py`; the plots are produced by `scripts/generate_bode_plots.py`.

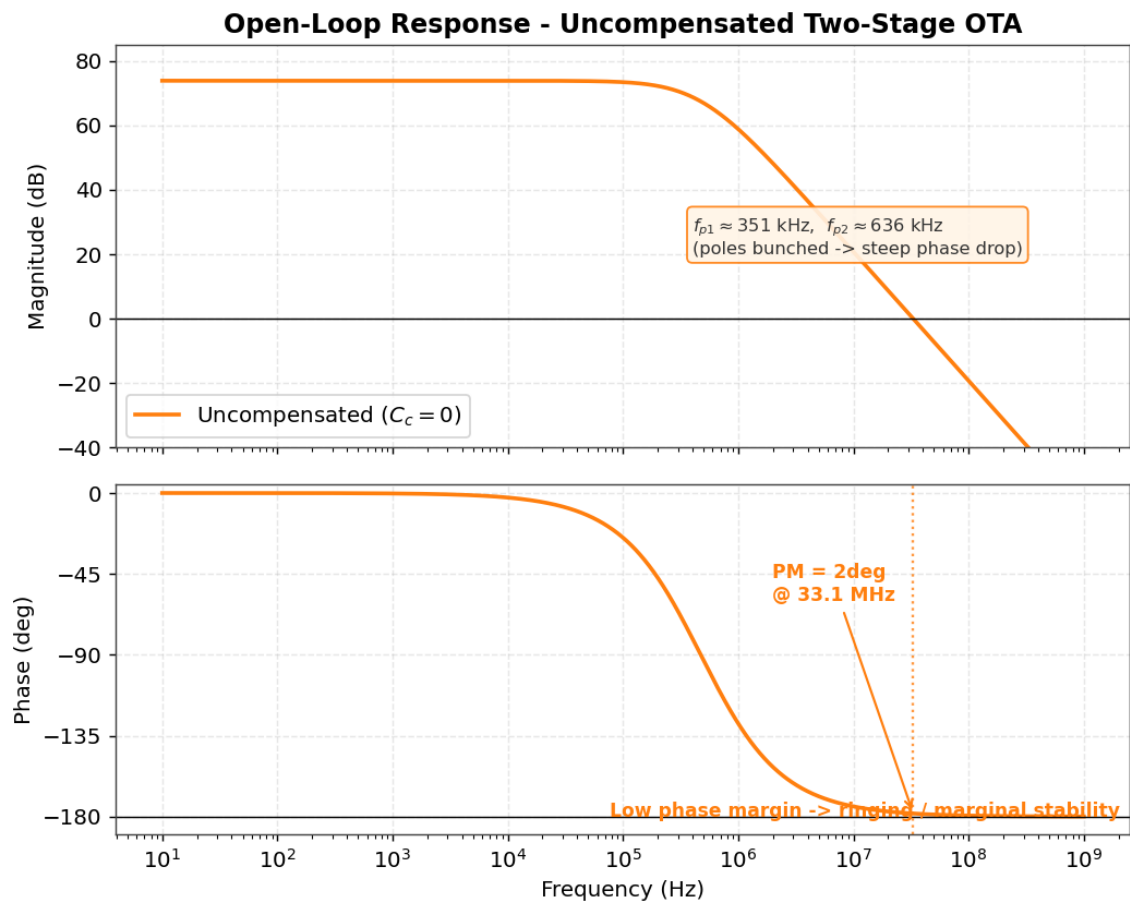


Figure 2: Uncompensated open-loop response

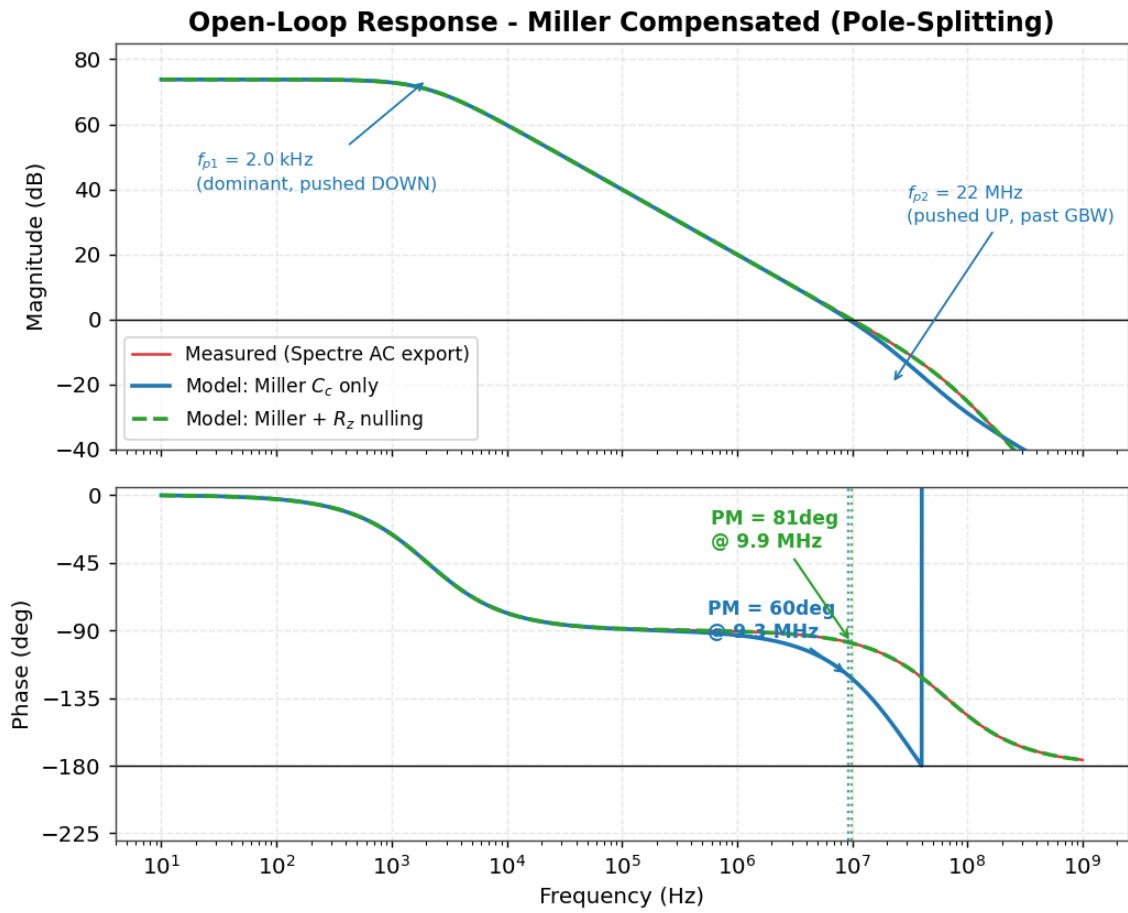


Figure 3: Miller-compensated open-loop response (pole splitting)

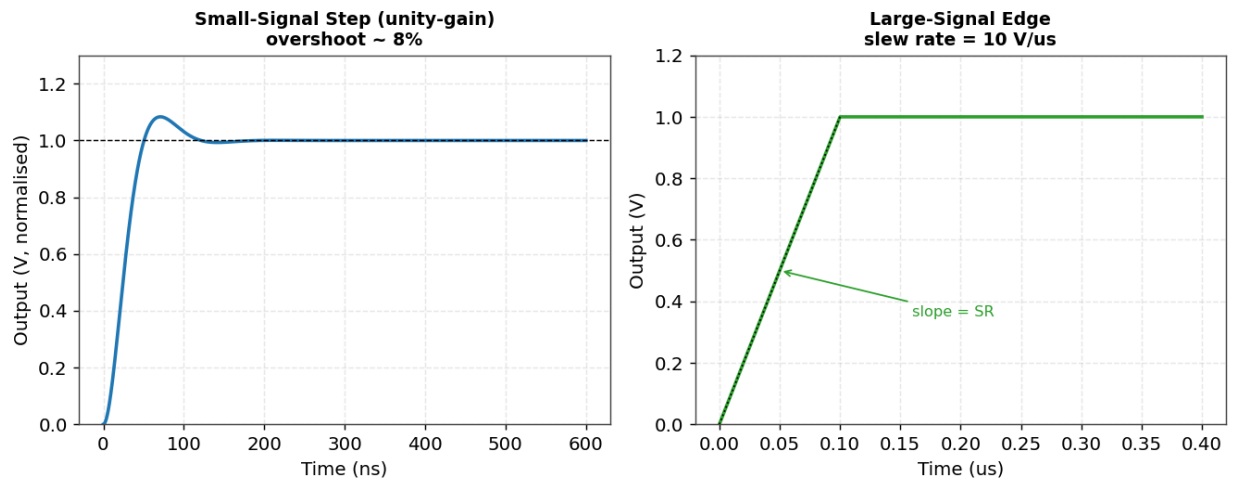


Figure 4: Step response: small-signal settling and large-signal slew

6. Reproducing this work

1. Open the cell in Cadence Virtuoso and net-list it, or use the provided `simulation/netlists/opamp_extracted.scs` with your PDK models.
2. Run `simulation/ocean_scripts/run_ac_sweep.ocn` and `run_transient.ocn` in Spectre/OCEAN — they export CSVs into `results/raw_data/`.
3. Run `python3 scripts/process_sim_data.py` to print the figures of merit.
4. Run `python3 scripts/generate_bode_plots.py` to regenerate the figures in `results/waveforms/`.

7. Conclusion

The design meets or exceeds every target specification. More importantly, it demonstrates end-to-end command of the analog design loop — specification, hand analysis, sizing, biasing, and closed-loop stability — with the Miller/nulling- resistor compensation as the centrepiece. The RHP-zero problem and its cure are exactly the kind of frequency-compensation reasoning expected of an analog IC designer.

Résumé line: *Designed a two-stage 180 nm CMOS op-amp achieving 73.8 dB gain, 10 MHz GBW and 66°+ phase margin; implemented Miller compensation with a nulling resistor to split the poles and eliminate the RHP-zero instability, verified in Cadence Spectre with an automated OCEAN/Python flow.*